

OLIVER AGDM, AB, Canada

WMO: 713510

Lat: 53.6466N

Lon: 113.3546W

Elev: 665

StdP: 93.59

Time Zone: -7.00 (NAM)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-32.2	-28.2	-35.7	0.1	-32.1	-31.5	0.2	-28.0	8.4	-2.8	7.4	-2.9	1.0	340	0.604

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.7	28.6	17.8	26.8	16.8	25.2	16.1	19.4	26.3	18.2	24.6	17.1	23.2	2.6	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
16.9	13.1	22.2	15.7	12.1	20.9	14.6	11.3	19.7	58.1	26.3	54.2	24.8	50.6	23.2	24.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
7.5	6.3	5.4	DB	-37.6	32.1	4.1	2.0	-40.5	33.5	-42.9	34.7	-45.2	35.8	-48.1	37.3
			WB	-37.7	21.5	4.0	1.2	-40.6	22.4	-43.0	23.1	-45.3	23.8	-48.2	24.7

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	3.2	-10.3	-10.2	-4.0	4.0	10.7	14.8	17.2	15.7	10.7	4.2	-4.3	-10.8	(6)
	DBStd	11.91	8.82	8.30	7.61	5.27	4.42	2.93	2.81	3.17	4.39	4.72	7.24	8.14	(7)
	HDD10.0	3164	629	564	433	187	44	2	0	1	43	186	429	644	(8)
	HDD18.3	5552	887	798	692	429	237	111	56	92	232	438	679	902	(9)
	CDD10.0	691	0	0	0	8	67	147	223	177	63	7	0	0	(10)
	CDD18.3	39	0	0	0	0	2	5	21	11	2	0	0	0	(11)
	CDH23.3	823	0	0	0	2	66	115	341	217	78	5	0	0	(12)
	CDH26.7	177	0	0	0	0	10	16	84	47	20	0	0	0	(13)
	WSAvg	2.2	2.1	2.0	2.2	2.8	2.7	2.4	2.1	2.0	2.1	2.3	2.1	1.9	(14)
(15) Precipitation	PrecAvg	420	18	12	16	27	49	69	90	54	27	20	15	17	(15)
	PrecMax	554	84	30	47	47	100	146	145	114	77	54	34	34	(16)
	PrecMin	235	3	5	5	3	8	20	19	17	1	4	3	5	(17)
	PrecStd	80	17	6	9	13	24	31	31	25	19	13	8	8	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.0	9.3	13.5	23.0	28.0	28.7	31.6	30.4	29.5	23.4	13.2	7.7	(19)
		MCWB	3.4	4.0	6.5	11.1	14.0	17.0	19.8	18.4	16.5	12.5	6.8	3.4	(20)
	2%	DB	5.9	6.1	10.3	19.4	25.2	26.4	28.9	27.7	25.4	17.7	9.2	5.1	(21)
		MCWB	2.4	2.0	4.6	9.1	12.7	15.8	19.0	17.8	14.7	10.1	4.5	1.8	(22)
	5%	DB	4.2	3.9	7.9	15.8	22.7	24.4	26.9	25.7	22.4	14.7	6.6	3.2	(23)
		MCWB	1.2	0.7	3.5	7.4	11.8	14.9	18.0	17.1	13.9	8.0	2.9	0.4	(24)
	10%	DB	2.0	1.9	5.7	13.3	20.3	22.4	25.0	23.8	19.3	12.2	4.6	1.0	(25)
		MCWB	-0.2	-0.5	2.0	6.1	10.7	14.1	17.1	16.3	12.3	6.7	1.8	-1.2	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.8	4.3	6.7	11.3	15.0	18.8	21.7	20.1	17.2	13.0	7.3	3.8	(27)
		MCDB	7.2	8.8	12.0	22.1	24.6	26.0	28.0	27.0	26.4	22.5	11.9	7.0	(28)
	2%	WB	2.6	2.2	5.0	9.5	13.6	17.1	20.1	18.9	15.6	10.5	5.0	2.0	(29)
		MCDB	5.6	5.5	9.8	18.2	22.1	23.6	27.2	25.9	24.1	17.1	8.4	4.6	(30)
	5%	WB	1.4	0.9	3.6	7.9	12.5	16.0	18.8	17.8	14.3	8.6	3.4	0.5	(31)
		MCDB	4.0	3.6	7.6	14.8	20.9	22.3	25.3	24.1	21.4	13.6	6.1	3.0	(32)
	10%	WB	-0.1	-0.4	2.2	6.5	11.4	15.0	17.6	16.7	12.9	7.2	2.0	-1.2	(33)
		MCDB	2.2	1.9	5.3	12.5	19.1	20.9	23.6	22.9	18.3	11.4	4.3	0.9	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	11.2	12.5	11.7	12.0	13.6	12.2	12.7	13.3	13.3	11.7	9.6	11.0	(35)
		MCDBR	11.1	14.2	13.7	17.3	18.1	16.1	16.6	17.2	19.3	16.7	11.6	11.4	(36)
	5% WB	MCWBR	8.2	10.5	8.8	8.9	8.1	7.1	7.8	8.3	9.7	9.4	7.7	8.7	(37)
		MCWBR	11.0	13.8	13.2	16.0	15.8	13.6	14.8	15.4	17.7	15.6	11.0	11.5	(38)
(40) Clear-Sky Solar Irradiance	taub		0.257	0.266	0.313	0.327	0.343	0.353	0.355	0.362	0.307	0.281	0.254	0.240	(40)
		taud	2.304	2.354	2.297	2.388	2.391	2.377	2.416	2.349	2.501	2.547	2.455	2.310	(41)
	Ebn at Noon		729	851	870	906	905	895	886	856	876	833	748	681	(42)
		Edh at Noon	57	75	100	105	110	113	107	108	81	62	49	46	(43)
(44) All-Sky Solar Radiation	RadAvg		0.96	2.00	3.55	4.67	5.73	6.02	6.15	5.06	3.49	2.00	0.96	0.71	(44)
	RadStd		0.11	0.15	0.27	0.44	0.42	0.39	0.30	0.35	0.36	0.24	0.11	0.07	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (69 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page